

thrust employed during the adjustment will reach the intended tissues in the VSC (Options, 2008). Appropriate stabilization without excessive restraint seems to reassure the patient, allowing them to relax into the adjustment, making the chiropractic treatment as effective as possible. Limb adjustments are accomplished by stabilizing the patient on three legs so that the chiropractor can isolate one joint at a time for adjustment. Many animal chiropractic providers report anecdotally that canine patients who are treated successfully with chiropractic adjustment appear to “enjoy” their subsequent experiences. It is a common occurrence to witness dogs straining at the leash in order to enter the facility for their chiropractic care.

Equine chiropractic adjustment technique varies depending upon the location of the adjustment. The cervical spinal segments are adjusted using a technique in which the horse’s head is positioned with one hand and the adjustment is performed with the other hand.

The thoracic and lumbar spinal segments and the pelvis are adjusted using a two-handed technique, and depending upon the location and direction of the adjustment, the technician may be asked to provide stabilization on the opposite side of the horse from the practitioner.

Horse limb adjustments are performed by lifting the leg, and isolating and adjusting one joint at a time.

Common sense will dictate which patients or locations on the body should not be treated with chiropractic adjustment. Examples of conditions that should not be treated with chiropractic include (but are not limited to):

- Areas with active infection
- Fractures
- Acute prolapsed or ruptured intervertebral disc
- Significant pain
- Joint luxation (e.g., traumatic coxofemoral luxation)
- Acute painful joint sprain or strain
- Meningitis/encephalitis.

The veterinary technician’s involvement with animal chiropractic cases will depend

upon the practice, the practitioner’s preferences, and the specific case. In those practices where the technician has first contact with the client and patient, it will be the technician who takes the preliminary history and conducts the initial pain assessment, reporting the findings to the chiropractor before the adjustment begins. During the adjustment, the technician provides restraint of the chiropractic patient as needed, as well as stabilization for specific adjustments.

For equine chiropractic patients, the technician will typically be positioned at the head on the same side of the horse as the chiropractor when the adjustment is along the spine or over the pelvis (Figure 24.1).

For adjustments in the thoracic spine of the horse, the technician will provide stabilization against the dorsal spinous processes on the side opposite to the chiropractor. The precise positioning of the technician’s hands, posture, and body will vary by practitioner. It is always appropriate to ask for guidance about body position when assisting with an equine chiropractic adjustment.



Figure 24.1 Dr. Downing adjusting an equine pelvis.



Figure 24.2 Dr. Downing performing adjustment with minimal restraint.

For canine chiropractic patients, the veterinary technician's role will vary depending upon the size of the patient (Figure 24.2).

For larger dogs, the technician will provide appropriate restraint to allow the chiropractor to isolate the segment being adjusting. During thoracic adjustments, in addition to restraining the patient, the technician will stabilize the appropriate ribs by placing the flat of the hand on the side of the ribcage opposite to the adjustment (Figure 24.3).

For limb adjustments, the technician will help to balance the dog while the chiropractor isolates and adjusts each joint. Finally, the technician will typically stabilize C1 during an occiput–C1 adjustment.

Because each practitioner has his or her own subtle variations of hand, body, and patient position during chiropractic adjustment, it is always best to seek direction in order to provide optimal assistance.

Chiropractic and Cats

Although current training of veterinarians and chiropractors in animal chiropractic principles and techniques focuses on horses



Figure 24.3 Hand position for stabilizing the thoracics during an adjustment.

and dogs, all quadrupeds share many structural, functional, and biomechanical commonalities. Cats should therefore benefit from the application of chiropractic adjustments within the context of physical rehabilitation when they experience VSCs. Feline patients are notorious for “masking” painful conditions, making pain recognition and subsequent treatment quite challenging.

The same principles apply to diagnosing and adjusting cats using chiropractic as apply to dogs, including decreasing significant pain before adjusting in order to provide the best outcome and experience for the patient. It also means appropriate gentle handling and minimal restraint to keep the patient, assistants, and the practitioner safe. Cats may object more vigorously to restraint than the typical dog. Likewise, cats tend not to stand still for their assessments and adjustments. Cats generally require flexibility on the part of the practitioner in order to accomplish appropriate chiropractic diagnosis and treatment (Figure 24.4).



Figure 24.4 Adjusting the patient when the table can be used to stabilize the patient.

Because of their small size, the veterinary technician's role during feline chiropractic adjustment may consist primarily of distracting the patient to minimize any objections it may have to being handled and manipulated.

Conclusion

The need for animal patient chiropractic studies is clear. Chiropractic care provides a reasonable strategy for complementing and enhancing the effects of pharmacological (and other) pain management strategies, and restoring biomechanically sound movement and function. Physical rehabilitation is all about restoring movement, function, strength, and ability. The future position of

chiropractic among accepted treatment modalities for animals undergoing physical rehabilitation will depend upon the results of rigorous clinical studies. Attention must be paid to the neurologic implications of chiropractic adjustment as well as sustaining the effects of adjustment.

Although it is important for veterinary technicians to understand the principles and practices of veterinary chiropractic as it can be applied within physical rehabilitation, it is not appropriate for technicians to perform animal chiropractic adjustments. The veterinary technician is positioned to play a vital role in providing chiropractic care to companion animals both large and small by working hand-in-hand with those practitioners who bring the benefits of chiropractic adjustment to their patients.

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25

Adjunctive Therapies Part 3: Myofascial Trigger Point Therapy

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Introduction

Myofascial pain syndrome, described in humans as a chronic painful condition of muscle, has started to become acknowledged in the field of veterinary medicine. Muscle is the largest organ in a mammal’s body (ranging from 40% in a human to 57% in a greyhound) and can be thought of as the “orphan organ,” as no traditional medical specialty is designated to treating it. Until recently, it was rare to provide veterinary education on muscle dysfunction, concentrating more on muscle tears and on the bones, joints, and ligaments of the musculoskeletal system. Although currently there is limited research on the development, diagnosis, and treatment of myofascial trigger points (MTrPs) in the non-human animal species, great strides

are being undertaken to transpose the human techniques initially developed by doctors into the realm of veterinary patients. With the development of the American College of Veterinary Sports Medicine and Rehabilitation and their mission “advances the art and science of veterinary medicine by promoting expertise in the structural, physiological, medical and surgical needs of athletic animals and the restoration of normal form and function after injury or illness” (ACVMSR, 2015), anticipation of advancements in the study of muscle pain and dysfunction exists.

Myofascial pain syndrome (MPS), is typically thought to arise from MTrPs. The pioneers recognizing MPS were Janet Travell MD and David Simons MD, and their textbook defines an MTrP as a hyper-irritable spot in skeletal muscle that is associated with

a hypersensitive palpable nodule in a taut band. The spot is tender when pressed, and gives rise to characteristic referred pain, motor dysfunction, and autonomic phenomena (Simons and Travell, 1998). The three components that make up an MTrP are the sensory, motor, and autonomic components.

Sensory Component

MTrPs can be very painful. The pain begins as nociception in the peripheral tissues from all the normal chemical activators, including serotonin, prostaglandins, bradykinin, and substance P. It is not within the scope of this chapter to go into details about all the pain-generating substances. These substances can decrease the activation threshold of a neuron, so that the nociceptor fires more easily with less of a stimulus, potentially leading to peripheral sensitization. Sampling of tissue fluid from active MTrPs in the upper trapezius of human patients was performed. Elevated levels of protons, bradykinin, serotonin, substance P, norepinephrine, calcitonin gene-related peptide, tumor necrosis factor alpha, and interleukin-1b were detected and were significantly different compared to samples taken from areas without MTrPs (Shah *et al.*, 2005). A persistent barrage of nociceptive signals from MTrPs may eventually lead to central sensitization, a form of neural plasticity involving functional, and/or structural change within the dorsal horn of the spinal cord. Central sensitization can be clinically expressed as allodynia (pain associated with a typically non-painful stimuli) or hyperalgesia (when an actual painful stimulus is perceived as more painful than it should).

Motor Component

MTrPs are also known as “contracture knots” due to the severe focal muscle contracture in the region of the MTrP. Contracture of the muscle fibers can compress local sensory

neurons as well as compressing local blood vessels, leading to a reduced supply of oxygen. MTrPs may inhibit normal motor activity in their muscle of origin or in functionally related muscles. Motor inhibition is often identified as muscle weakness and can lead to poor coordination and muscles imbalances (McPartland and Simons, 2011). Clinically this can appear as muscle weakness without atrophy and unrelated to neurologic causes (Simons and Travell, 1998).

Autonomic Component

The autonomic nervous system (ANS) primarily exerts control over cardiac tissue, blood vessels, visceral organs, and glands of the body. Typically, ANS responses seen in human patients with MTrPs include localized sweating, vasoconstriction or vasodilation, and pilomotor activity (goose bumps) (McPartland and Simons, 2011). Although these responses are seen frequently and are easily detectable in humans, the detection of any ANS response is rare in the veterinary patient – with pilomotor activity (piloerection) being the most likely event noted.

How Do Myofascial Trigger Points Develop

Several possible mechanisms can lead to the development of MTrPs, including low-level muscle contractions, uneven intramuscular pressure distribution, direct trauma, eccentric contraction in unconditioned muscle, and maximal or submaximal concentric contractions. The exact reason as to why these mechanisms lead to the formation of MTrPs remains unclear. The original hypothesis introduced in 1981 as the integrated trigger point hypothesis was dependent on an energy crisis that would result in excessive acetylcholine at the motor endplate. The excess of acetylcholine would then cause contraction of the sarcomere leading to the development

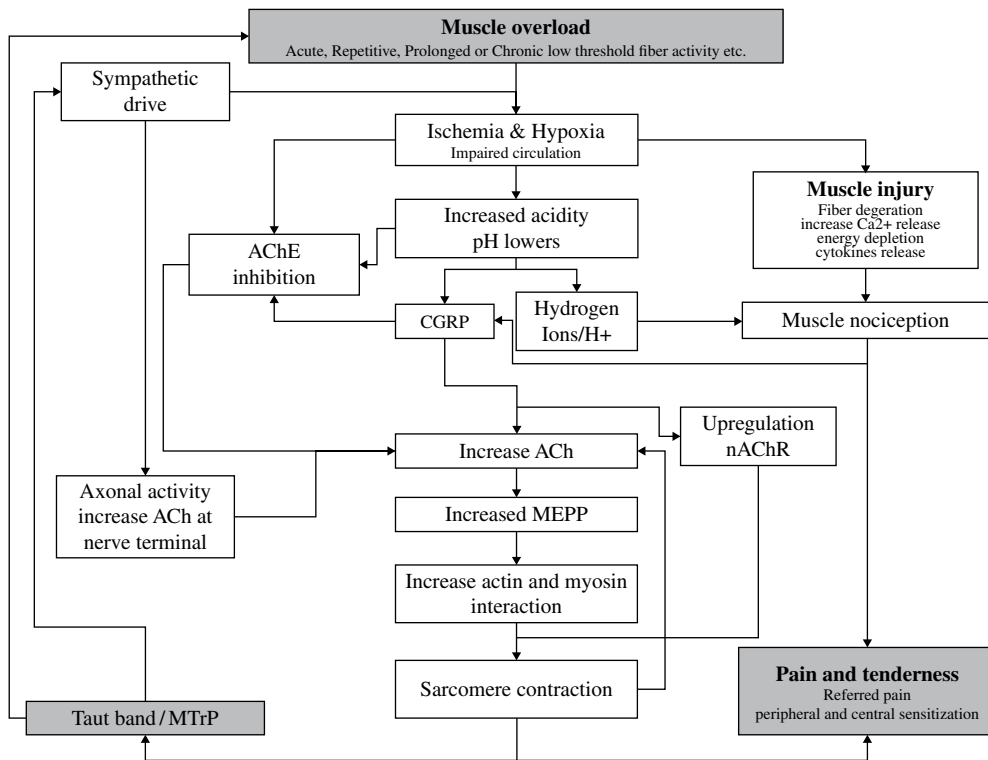


Figure 25.1 Schematic showing the expanded integrated trigger point hypothesis. *Source:* Courtesy of Dr. Jan Dommerholt.

of MTrPs. With advances in research, this hypothesis became the expanded integrated trigger point hypothesis in 2004 (Gerwin *et al.*, 2004) (Figure 25.1).

Low-level muscle contractions can result in degeneration of muscle fibers. This can lead to an increase in calcium release, energy depletion, and structural damage especially to the mitochondria. During low-level muscle contractions, intramuscular pressure can lead to excessive capillary pressure causing localized hypoxia and ischemia. Direct trauma may create a vicious cycle of events wherein damage to the sarcoplasmic reticulum or the muscle cell membrane may lead to an increase of calcium concentration, a subsequent activation of actin and myosin, a relative shortage of adenosine triphosphate (ATP), and an impaired calcium pump, which in turn will increase the intracellular calcium concentration even more, completing the cycle.

Eccentric and concentric exercises have been associated with hypoxia and ischemia (see Figure 25.1). However, there is currently inadequate evidence to demonstrate that these exercises are absolute precursors to the development of MTrPs (Dommerholt *et al.*, 2011; Bron and Dommerholt, 2012).

Perpetuating Factors

Perpetuation of MTrP formation in dogs appears to be most often related to mechanical stresses resulting in chronic muscle overload. Postural changes in dogs resulting from orthopedic injury, postoperative surgical trauma and pain, neuropathy, joint dysfunction, and pain related to osteoarthritis create muscle overload. Many of the same muscle-related mechanisms that lead to the development of MTrPs are also perpetuating factors.

An example to illustrate this point is the canine patient with chronic osteoarthritis of the coxofemoral joints. The pain caused by the joints will cause the patient to invoke compensatory postural changes to relieve pressure on the joints, thereby activating and perpetuating MTrPs. MTrPs will readily be found in the hip flexors (e.g., iliopsoas, tensor fasciae latae, sartorius, and rectus femoris), extensors (gluteals, piriformis, semimembranosus, and semitendinosus) and adductors (gracilis, pectineus, and adductor). As the patient shifts weight cranially, thereby changing the typical 60/40 weight distribution pattern, compensatory changes will be found in the muscles of the forelimbs. The overload placed on the forelimbs can lead to the formation and presence of MTrPs; primarily in infraspinatus, triceps, teres major, and deltoids. As the patient ambulates, there will be considerable lateral flexion of the spinal muscles to avoid extensive coxofemoral flexion and extension. This can be described as a “hula type motion” when viewed from the caudal aspect of the patient. This overuse of the iliocostalis lumborum can result in the formation of MTrPs. This example of formation of trigger points secondary to joint pain not only illustrates the perpetuation that can be involved with MTrPs, but it also demonstrates the need to treat any underlying pain. If the MTrPs are cleared with treatment and any known underlying source of pain is not addressed, then the MTrPs will return and the patient will continue to suffer from myalgia as well as the underlying pain. This is just one illustration of a condition that can perpetuate MTrPs; similar findings can be found with patients suffering from shoulder injuries, antebrachial injuries, chronic medial luxating patella, and cruciate injuries among others.

In humans, MTrPs have been found from nutritional, metabolic, or systemic perpetuating factors, including the use of statin-class drugs, iron insufficiency, vitamin D insufficiency, vitamin B12 (cobalamin) insufficiency, and hormonal imbalances, such as is seen with hypothyroidism (Dommerholt and

Gerwin, 2011). Hypothyroidism is the most common endocrine disorder in dogs and is associated with a variety of clinical signs; however, the veterinary literature does not mention pain resulting from hypothyroidism (Wall, 2014). However, one might pause to consider those hypothyroid dogs that suffer from myopathy and muscle weakness and closely evaluate the patient for MTrPs.

Evaluation of the Patient

Close observation of the patient prior to placing your hands on them can provide you with a lot of information and knowledge. Observing the patient at various gaits (walk, trot, jog) or the unwillingness to move; observing position changes such as stand to sit or laying down to stand can provide the astute observer with considerable information as to which muscle or muscle groups may be involved. Video-taping the patient to analyze in slow motion can also be very beneficial. In dogs, a simple test can be performed to evaluate muscle weakness. With a dog in standing position, slowly slide the limb backward until non-weight bearing. A slight to profound drop of the contralateral side can be indicative of muscle weakness or altered muscle firing patterns associated with MTrPs within the antigravity muscles of that limb (Wall, 2014).

MTrPs are classified as active or latent. Active MTrPs cause local or referred pain patterns while at rest, with muscle movement or upon direct stimulation. Latent MTrPs do not trigger pain without direct stimulation (Dommerholt *et al.*, 2011). Both active and latent MTrPs are painful on compression, thereby making it virtually impossible for the veterinary examiner to know if a patient is suffering from active or latent MTrPs, so all MTrPs that are found should be addressed as a possible cause of pain and muscle dysfunction (Figure 25.2). Identification of taut bands and hypersensitive MTrPs within muscle is an acquired skill set that requires an understanding of these

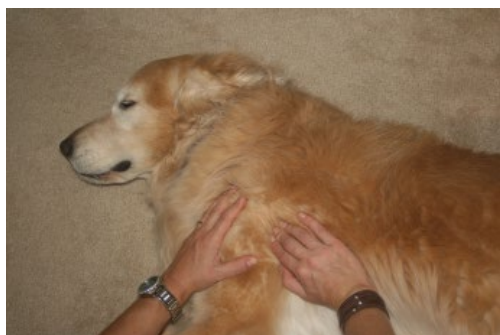


Figure 25.2 Example of flat palpation in a dog for the presence of myofascial trigger points.



Figure 25.3 Example of flat palpation on a horse for the presence of myofascial trigger points.

changes, skilled instruction, and repeated practice (Wall, 2014).

Without a “gold standard” of laboratory testing or diagnostic imaging to diagnose MTrPs we are heavily reliant on palpation skills. Palpation for MTrPs can be accomplished by two simple techniques. Flat palpation is the gentle but firm movement of the finger tips perpendicular to the muscle fibers being examined as they are pushed against the underlying bone (Figure 25.3). This technique lends itself to examination of muscles such as the iliopsoas, infraspinatus, and supraspinatus.

Pincer palpation is performed by rolling the muscle perpendicular to the muscle fibers between the finger tips and thumb. This technique lends itself to the examination of muscles such as the sartorius, tensor fasciae latae, and triceps (Figure 25.4). Once a taut band is detected, the taut band is evaluated for an area of hardness and discrete pain.

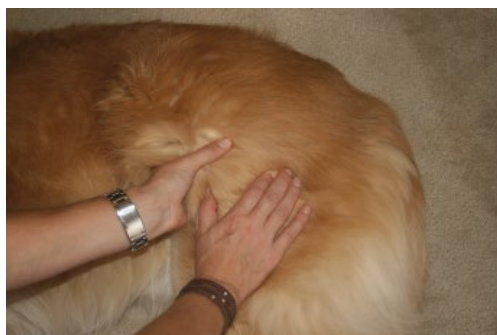


Figure 25.4 Pincer palpation on a dog.

As the stimulation of an active or latent MTrP can lead to a “jump sign,” a pain response leading to vocalization or the patient trying to move away, palpation is best done with an assistant providing gentle restraint. Although palpation can be performed on a standing patient, many people find it is much easier to palpate the taut bands in a relaxed muscle in a non-weight bearing lateral recumbent position.

A “local twitch response” (LTR) may be induced during the palpation over an MTrP. The LTR is a unique spinal cord reflex resulting in a rapid contraction of the taut band following manual stimulation of the MTrP. Manual stimulation can be accomplished by direct palpation or introduction of a needle. The LTR in dogs can also serve as verification of the presence of an MTrP (Wall, 2014).

The criteria for identification of MTrPs can be best summed up by: a palpable taut band, exquisite spot tenderness of a nodule in the taut band, and patient recognition of pain upon palpation of the nodule (jump sign in the veterinary patient). Confirmatory observations in the veterinary patient would be a visual or tactile identification of an LTR (Simons and Travell, 1998).

Treatment of Myofascial Trigger Points

Therapies fall into two categories: invasive and non-invasive. At this time, no clinical studies have been done to evaluate the outcome

or validate the effectiveness of any treatment for MTrPs in the veterinary patient. All reports are considered anecdotal and symptomatic and clinical improvement or failure of treatment is based strictly on “observation” by the clinician and the owner.

Non-Invasive Therapies

Electrotherapy

Anecdotal reports from several veterinary clinicians report successful treatment of MTrPs with the Pointer Excel II™, a handheld transcutaneous electrical nerve stimulation device. The Pointer Excel II™ is placed on the skin over the MTrP and electrical intensity is increased until rhythmic muscle contraction is produced. Response to therapy is likely related to rapid muscle stretch that occurs with induced contraction rather than the introduction of electrotherapy (Wall, 2014).

Low-Level Laser Therapy

This chapter will only discuss the use of class IIIa and IIIb lasers. As the class IV laser (output power of greater than 500mW) must continually be moved over the surface of the skin to prevent thermal damage, it is difficult to determine the exact J/cm² that would be delivered to the underlying MTrP. Class IIIa lasers produce an output power of up to 5mW; whereas Class IIIb lasers produce an output power up to 500mW. During laser therapy the energy delivered is reported in joules (J), where 1 J = 1 W/s and the treatment area is described in centimeters squared, therefore the therapeutic laser dose is described as J/cm². Although many research studies have investigated low-level laser therapy for the treatment of MTrPs, there is no conclusive evidence to support or deny its effectiveness (Ilbuldu *et al.*, 2004; Altan *et al.*, 2005; Dundar *et al.*, 2007; Manca *et al.*, 2014). Inadequate dosages may be the principal factor involved in the inconsistency among reports of LLLT efficacy. Until there is further research to demonstrate an effective therapeutic laser dose for the treatment of MTrPs, this modality should not be a standalone treatment procedure.

Ultrasound

Current research demonstrates that the use of conventional therapeutic ultrasound is no more effective than placebo or no treatment for MTrP pain in the neck and upper back (Lee *et al.*, 1997; Gam *et al.*, 1998; Esenyel *et al.*, 2000; Manca *et al.*, 2014).

Shockwave Therapy

Current research, although limited in number of studies, is demonstrating some effectiveness for the treatment of MTrPs and this modality will need further research to fully elucidate its role as a therapy (Jeon *et al.*, 2012; Ji *et al.*, 2012; Moghtaderi *et al.*, 2014). It is the clinical observation of the authors that after shockwave therapy is performed on a patient with osteoarthritis of the coxofemoral joint, the number of palpable MTrPs in the tensor fasciae latae, proximal sartorius, and gluteals muscles is less than that before the therapy is done.

Manual Therapies

In humans, the evidence for manual therapies is once again not clear, as the data regarding most manual therapies is inadequate and conflicting. Most trials do not limit the number of modalities per treatment, so positive outcomes cannot be exclusively claimed by a certain therapy. Several studies reported that exercise and stretching appeared to be the effective therapy when included in treatment groups comparing active and placebo modalities (Mense and Gerwin, 2010; Rickards, 2011). Research on trigger point pressure release has shown positive outcomes as a treatment in human subjects. This technique is one of the more commonly described manual techniques to address MTrPs (Bodes-Pardo *et al.*, 2013; Cagnie *et al.*, 2013; Llamas-Ramos *et al.*, 2014). Trigger point release consists of pressure progressively applied and increased over the MTrP until the fingers encounter an increase in tissue resistance (tissue barrier). This pressure is maintained until the clinician senses a relief of the taut band. At that moment, the pressure is increased again until the next increase in tissue resistance is felt.

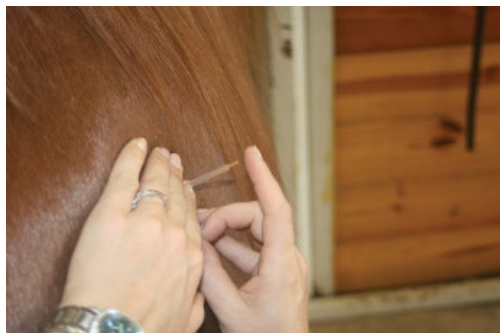


Figure 25.5 Dry needling of a horse for treatment of myofascial trigger points by Mrs. Angela Stramel.



Figure 25.6 Dry needling of a dog for treatment of myofascial trigger points by Dr. Douglas Stramel.

The process is repeated 3 times during each session (Lewit, 1991).

Invasive Therapies

Although veterinary technicians are not allowed to perform these procedures, a knowledge of the procedure and proper restraint or sedation for the procedure is important for technicians in their role as part of the animal healthcare team. Invasive procedures consist of dry needling (DN) or trigger point injections (TPI). TPI involves the use of a hypodermic needle attached to a syringe to inject a substance directly into the MTrP. Commonly used substances in the human world include lidocaine and botulinum toxin. Due to the discomfort from these injections, this procedure is not well tolerated in the veterinary patient. DN consists of the use of a filiform needle inserted into the skin and then into the deeper underlying muscle (Figure 25.5).

The MTrP within the taut band is targeted and eliciting LTRs is essential. Once an LTR is evoked, the needle is slightly withdrawn from the muscle, but not the skin, and can be redirected to the same MTrP until no further LTRs are evoked (Figure 25.6). Additional MTrPs in the area can also be treated prior to the needle being completely removed from the patient. The human literature covers in depth the advantages of DN (Dommerholt and Gerwin, 2010).

Treatment outcome goals would include a decrease in peripheral and central sensitization as well as improved motor function as seen with increase in muscle strength and range of motion leading to a better quality of life for the veterinary patient. A successful outcome of DN for the treatment of MTrPs is heavily reliant on the knowledge and skills of the veterinary clinician. One must have an in-depth knowledge of anatomy and develop keen tactile awareness and visualization of the needle pathway as it travels through the soft tissue of the patient's body (Dommerholt and Gerwin, 2010).

Conclusions

If you have an awareness of MTrPs affecting the veterinary patient, you can better assess those patients suffering from myalgia and having a decline in athletic performance. Knowledge that MTrPs can complicate, prolong, or delay the pain management and healing process in the chronic or acute orthopedic and neurologic patient can vastly change the way patients are treated. These patients have a better outcome than those left untreated. The skill set to be able to understand, palpate, and treat MTrPs is essential for each member of the veterinary healthcare team that is interested in pain management, rehabilitation, and sports medicine.

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